

Lecture 21.

$$\frac{d}{dx} \int_a^x f(t) dt = f(x). \quad (\text{FT(2.)})$$

Solve $y' = \frac{1}{x} \Rightarrow \boxed{L(x) = \int_1^x \frac{dt}{t}}$

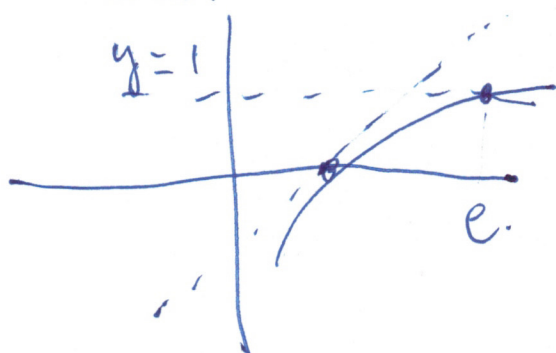
Definition of LOG

$$L'(x) = \frac{1}{x}; \quad L(1) = \int_1^1 \frac{dt}{t} = 0.$$

$$L''(x) = -\frac{1}{x^2} \quad (\text{concave down})$$

$$L(1) = 0 \quad L'(1) = \frac{1}{1}$$

Sketch



Definition

$$e = L(e) = 1$$

Why $L(x) < 0$ for $x < 1$

① $L(1) = 0$, and L is increasing

② $L(x) = \int_1^x \frac{dt}{t} = - \int_x^1 \frac{dt}{t}$

< 0

> 0 when $x < 1$

Claim: $L(ab) = L(a) + L(b)$

$$\int_1^{ab} \frac{dt}{t} = \int_1^a \frac{dt}{t} + \int_a^{ab} \frac{dt}{t}$$

$L(ab) \quad L(a)$

$$\int_a^{ab} \frac{dt}{t} = \int_{u=1}^{u=b} \frac{adu}{au} = \int_{u=1}^{u=b} \frac{du}{u} = L(b).$$

$$dt = a du.$$

$$t=a \rightarrow u=1$$

$$t=ab \rightarrow u=b$$

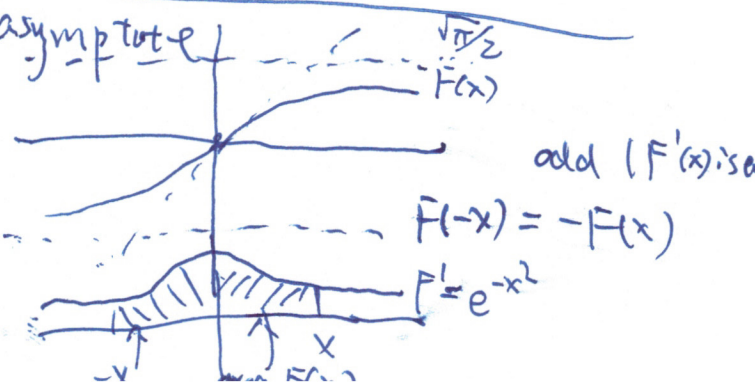
$$F(x) = \int_0^x e^{-t^2} dt$$

$$F'(x) = e^{-x^2}, \quad F(0) = 0.$$

$$F''(x) = -2xe^{-x^2} \begin{cases} < 0, x > 0 \\ > 0, x < 0. \end{cases}$$

$$F'(0) = e^{-0^2} = 1.$$

asymptote



$$\lim_{x \rightarrow \infty} F(x) = \frac{\sqrt{\pi}}{2}$$

$$\infty \quad -\frac{\sqrt{\pi}}{2}$$

$$\operatorname{erf}(x) = \frac{\pi}{\sqrt{2}} \int_0^x e^{-t^2} dt = \frac{2}{\sqrt{2}} F(x)$$

$$C(x) = \int_0^x \cos(t^2) dt$$

$$S(x) = \int_0^x \sin(t^2) dt$$

$$H(x) = \int_0^x \frac{\sin t}{t} dt$$

functions

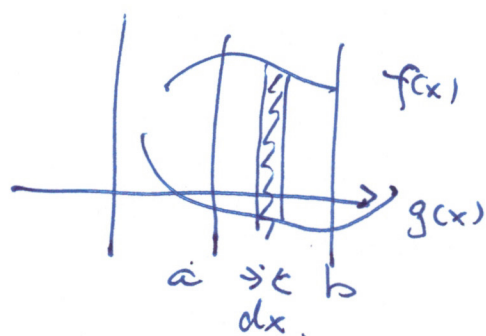
can only be expressed as

integration !!!

$$Li(x) = \int_2^x \frac{dt}{\ln t}$$

$$Li(x) \approx \# \text{ primes } < x$$

Area between curves

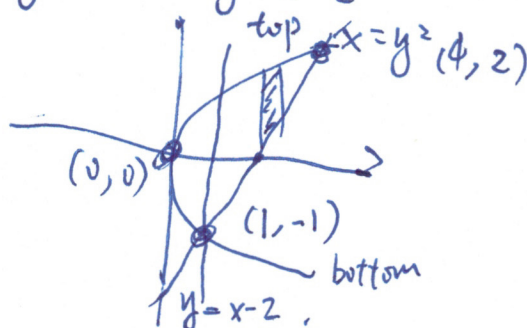


$$\text{Area} = \int_a^b (f(x) - g(x)) dx$$

Limits
I
Integral
base

Example. Find area between $x=y^2$ and $y=x-2$

Step 1. draw a picture.



1) find 3 points
 $x = y^2$ intersect $y = x - 2$

at

$$y = y^2 - 2 \Rightarrow y = 2 \text{ or } y = -1$$

$$\text{top} = y = \sqrt{x}$$

$$\text{bottom left} = y = -\sqrt{x}$$

$$\text{bottom right} = y = x - 2$$

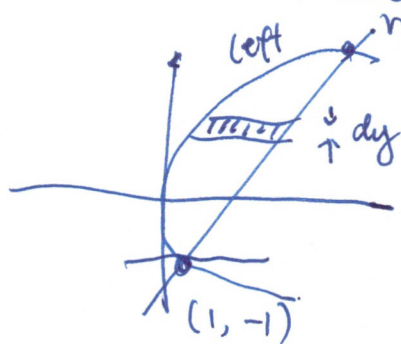
$$\text{Area} = \underbrace{\int_0^1 (\sqrt{x} - (-\sqrt{x})) dx}_{\text{left}} + \underbrace{\int_1^4 (\sqrt{x} - (x-2)) dx}_{\text{right}} = \dots$$

Method 1. (hard way)

right

Method 2. (better)

use horizontal slices.



$$\text{left } x = y^2$$

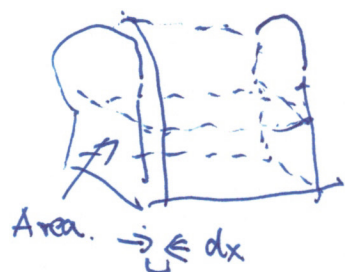
$$\text{right } y = x - 2 \Rightarrow x = y + 2$$

$$\text{Area} = \int_{-1}^2 (y+2 - y^2) dy = \left(\frac{y^2}{2} + 2y - \frac{y^3}{3} \right) \bigg|_{-1}^2 = 9/2$$

$$y+2 > y^2$$

$$y+2 - y^2 > 0$$

Lecture 22. Volumes by slicing.



slice of
breach

$$\Delta V = A \Delta x$$

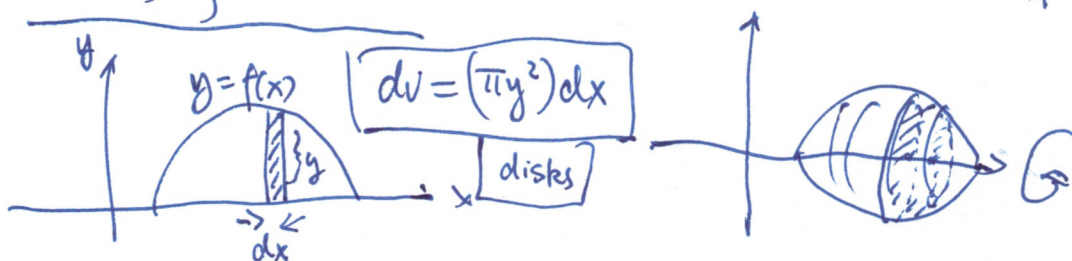
$$dV = A(x) dx$$

$$V = \int A(x) dx \approx \sum_i A_i \Delta x.$$

Solids of Revolution.

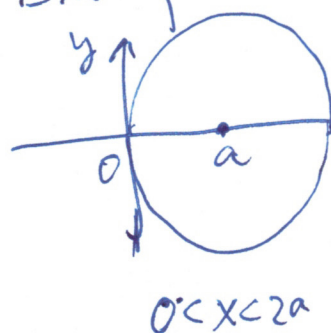
(only draw 2D!)

(Method 1 = method of disks)



revolve around x axis

Example. Vol of ball of radius a.



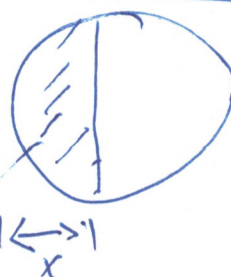
$$dV = \pi y^2 dx$$

$$V = \int_0^{2a} \pi (2ax - x^2) dx$$

$$(x-a)^2 + y^2 = a^2.$$

$$y^2 = a^2 - (x-a)^2 = 2ax - x^2.$$

$$\begin{aligned} V &= \pi \left(ax^2 - \frac{x^3}{3} \right) \Big|_0^{2a} \\ &= \pi \left(4a^3 - \frac{8a^3}{3} \right) \\ &= \boxed{\frac{4}{3} \pi a^3} \end{aligned}$$



$$Vol = \pi \left(ax^2 - \frac{x^3}{3} \right)$$

$V(x)$ = Vol of portion of width x of ball

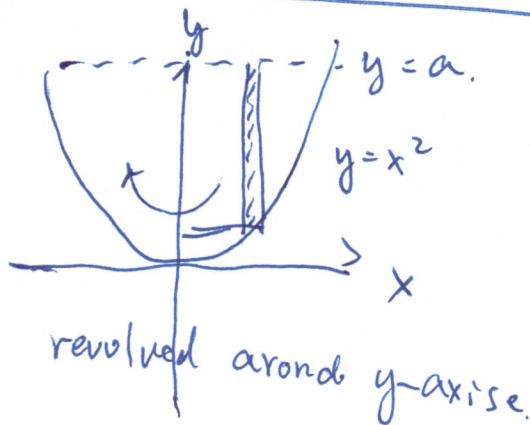
Double check: $V(k) = V$ of half ball

$$V(k) = \pi \left(a^3 - \frac{a^3}{3} \right) \\ = \pi \frac{2}{3} a^3 \checkmark$$

(Method¹² of Shells)

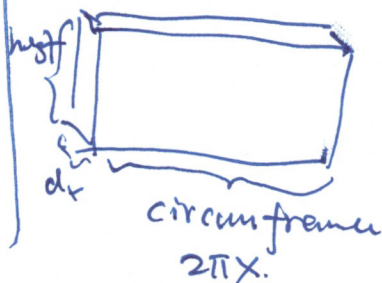


Witch's
Cauldron



thickness = dx
height = $a - y$ ($y = x^2$)
($y_{\text{top}} - y_{\text{bottom}}$)

unwrapped.



$$dv = (2\pi x)(a - y) dx \\ = 2\pi x(a - x^2) dx \\ = 2\pi(ax - x^3) dx \\ 0 < x < \sqrt{a}$$

$$V = \int_0^{\sqrt{a}} 2\pi(ax - x^3) dx \\ = 2\pi \left(\frac{ax^2}{2} - \frac{x^4}{4} \right) \bigg|_0^{\sqrt{a}} \\ = 2\pi \left(\frac{a^2}{2} - \frac{a^2}{4} \right) = \frac{\pi}{2} a^2.$$

Beware of units.

$$a = 100 \text{ cm}$$

$$V = \frac{\pi}{2} (100)^2 \text{ cm}^3$$

$$= \frac{\pi}{2} 10 (1000^3) = \underline{16 \text{ liter.}}$$

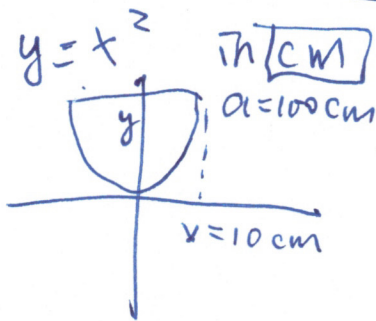
$$a = 1 \text{ M} \quad V = \frac{\pi}{2} \text{ m}^3$$

$$= \frac{\pi}{2} (100 \text{ cm})^3 = \frac{\pi}{2} 10^6 \text{ cm}^3$$

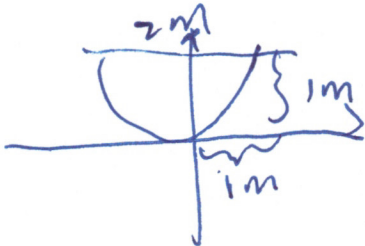
$$= \frac{\pi}{2} 1000 \text{ liters} = \underline{1600 \text{ liters}}$$

Both answers are correct!!!

$y = x^2$ can be interpreted
in two ways!!



$y = x^2$ in meters



Lecture 24.

AVERAGE VALUE

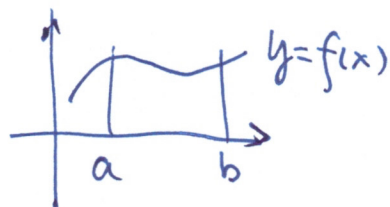
$$\frac{y_1 + \dots + y_n}{n}$$

$n \rightarrow \infty$

$$\boxed{\frac{1}{b-a} \int_a^b f(x) dx}$$

continuous
avg

AVG(f)



$$\Delta x = \frac{b-a}{n}$$

$$a \leq x_0 < x_1 < \dots < x_{n-1} < x_n = b$$

$$y_1 = f(x_1), y_2 = f(x_2), \dots, y_n = f(x_n)$$

RiemSum

$$\frac{(y_1 + \dots + y_n) \Delta x}{b-a} \xrightarrow{\Delta x \rightarrow 0} \frac{\int_a^b f(x) dx}{b-a}$$

$$\frac{\Delta x}{b-a} = \frac{1}{n} \rightarrow 0. (n \rightarrow \infty)$$

Ex 1. $f(x) = C$.

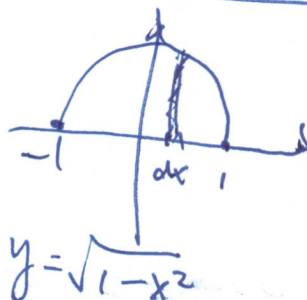
$$\frac{1}{b-a} \int_a^b C dx = C.$$

AVG(C) = C, explains $\frac{1}{b-a}$ (and $\frac{1}{n}$)

Ex 2. Average height of a point on a semi circle.

$$\frac{1}{+1 - (-1)} \int_{-1}^1 \sqrt{1-x^2} dx = \frac{1}{2} \left(\frac{\pi}{2} \right) = \frac{\pi}{4}$$

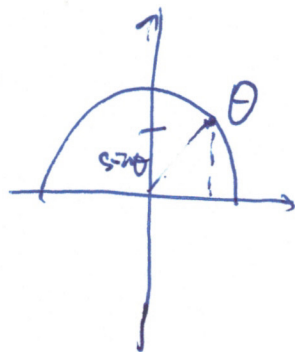
↑
area of a circle.
semi



Ex3. Avg ht with respect of y vs $\rightarrow$ with respect to x
 $(y = \sin \theta)$

arc length θ θ $(0 \leq \theta \leq \pi)$
 $(y = \sin \theta)$

$$\frac{2}{\pi} < \frac{\pi}{4}$$



$$\frac{1}{\pi} \int_0^{\pi} \sin \theta d\theta = -\frac{1}{\pi} \cos \theta \Big|_0^{\pi} = -\frac{1}{\pi} (\cos \pi - \cos 0) = \frac{2}{\pi}$$

WEIGHTED AVERAGE.

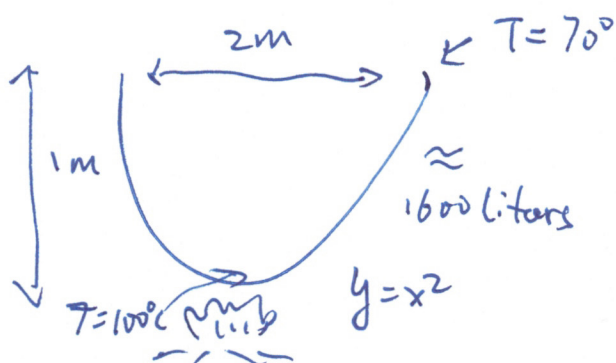
$\Rightarrow \text{Avg}(c) = c.$

$$\frac{\int_a^b f(x) w(x) dx}{\int_a^b w(x) dx}$$

$$\int_a^b c w(x) dx$$

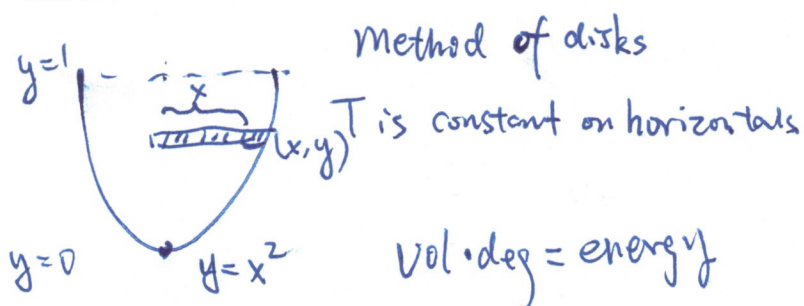
$$\frac{\int_a^b c w(x) dx}{\int_a^b w(x) dx} = c. \text{ \& explains denominator}$$

Witche's cauldron



Initial temp $T = 0^\circ \text{C}.$

Final temp $T = 100 - 30y$



Method of disks

T is constant on horizontals

Vol. $\cdot$ deg = energy

$$\int_0^1 T(\pi x^2) dy.$$

$$= \int_0^1 (100 - 30y)(\pi y) dy.$$

$$=$$

$$= 50\pi y^2 - 10\pi y^3 \Big|_0^1$$

$$= \boxed{40\pi} \text{ deg. m}^3$$

celcius

$$(40\pi) \text{ deg. m}^3 \left(\frac{1 \text{ cal}}{\text{deg. cm}^3} \right) \left(\frac{100 \text{ cm}}{\text{m}} \right)^3$$

$$= 40\pi \cdot 10^6 \text{ cal}$$

$$= 40\pi \cdot 1000 \text{ kcal}$$

Average (final) temperature.

$$\frac{\int_0^1 T \pi y dy}{\int_0^1 \pi y dy} = \frac{40\pi}{\pi/2} = 80.$$

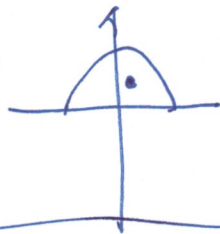
$w(y) = \pi y.$

$$\frac{T_{\max} + T_{\min}}{2} = 85. > 80.$$

Probability:

Pick a point "at random".

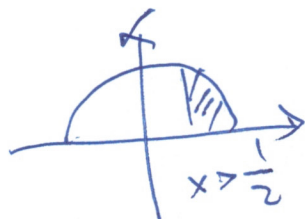
in $0 < y < 1 - x^2$.



what's the probability $x > \frac{1}{2}$

$$= P(x > \frac{1}{2}).$$

Probability = $\frac{\text{Part}}{\text{Whole}}$



$$\int_{\frac{1}{2}}^1 (1-x^2) dx$$

$$\frac{\int_{\frac{1}{2}}^1 (1-x^2) dx}{\int_{-1}^1 (1-x^2) dx} = P(x > \frac{1}{2}) = 5/18.$$

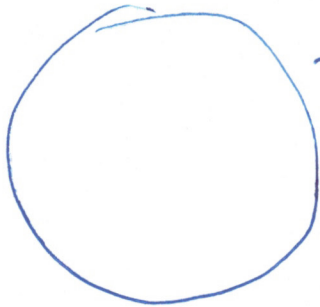
$$\int_{-1}^1 (1-x^2) dx$$

$$w(x) = 1 - x^2. \quad a = -1, b = 1.$$

General formula.

$$a \leq x_1 < x_2 \leq b$$

$$P(x_1 < x < x_2) = \frac{\int_{x_1}^{x_2} w(x) dx}{\int_a^b w(x) dx} = \frac{\text{Point}}{\text{Whole}}$$



target, throwing darts



Lecture 24.

Dart Board.

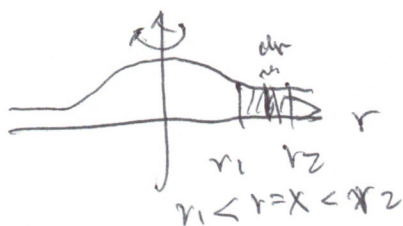


What is the probability that your little brother gets hit by a dart.

$$\# \text{ hits} = C e^{-r^2}$$



Sideview of e^{-r^2}

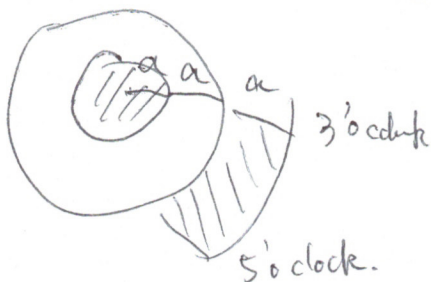


$$\begin{aligned} \text{Shells} &= \int_{r_1}^{r_2} 2\pi r e^{-r^2} dr \\ &= -\pi e^{-r^2} \Big|_{r_1}^{r_2} \\ &= \pi (e^{-r_1^2} - e^{-r_2^2}) \end{aligned}$$

$$\begin{aligned} \text{Part} &= C\pi (e^{-r_1^2} - e^{-r_2^2}) \\ \text{WHOLE} &= 0 \leq r < \infty \\ &= C\pi (e^{-0^2} - e^{-\infty^2}) \\ &= C\pi \end{aligned}$$

$$P(r_1 < r < r_2) = \frac{\text{Part}}{\text{Whole}} = \frac{e^{-r_1^2} - e^{-r_2^2}}{e^{-0^2} - e^{-\infty^2}}$$

Ralph: $P(0 \leq r < a) = 1/2$
 $a = \text{radius of target}$

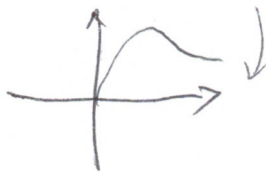


$$\begin{aligned} \frac{1}{6} P(2a < r < 3a) &\approx \frac{1}{6} \times \frac{1}{16} \times \frac{1}{2} \\ e^{-0^2} - e^{-a^2} &= 1/2 \\ e^{-a^2} &= 1/2. \end{aligned}$$

$$\begin{aligned} P(2a < r < 3a) &= e^{-(2a)^2} - e^{-(3a)^2} \\ &= (e^{-a^2})^4 - (e^{-a^2})^9 \\ &= \left(\frac{1}{2}\right)^4 - \left(\frac{1}{2}\right)^9 \approx \frac{1}{16} \end{aligned}$$

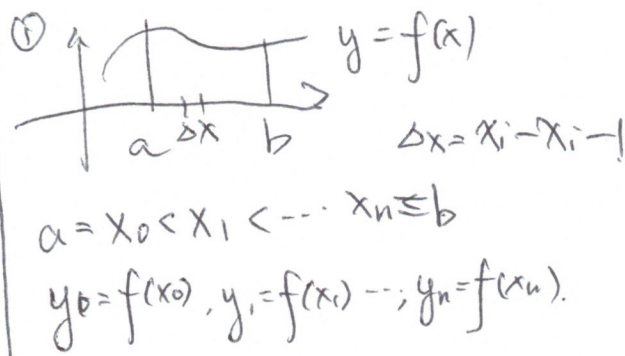
Weight

$$W(r) = 2\pi r e^{-r^2}$$



Numerical Integration.

1. Riemann Sums
2. trapezoidal rule
3. Simpson's rule

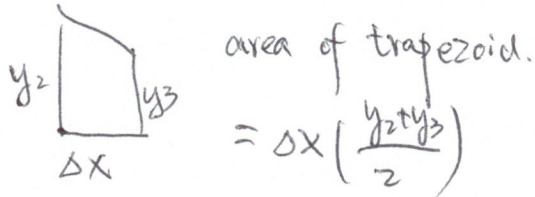


Goal: "average" or add up to get approximation to $\int_a^b f(x) dx.$

Riemann Sum

$$(y_0 + y_1 + \dots + y_{n-1}) \Delta x \quad (\text{left sum})$$

$$(y_1 + y_2 + \dots + y_n) \Delta x \quad (\text{right sum})$$

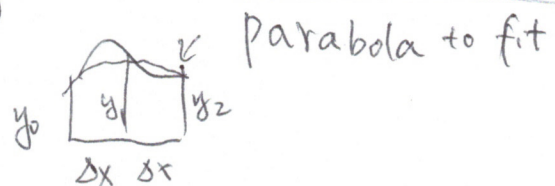


$$\Delta x \left(\frac{y_0 + y_1}{2} + \frac{y_1 + y_2}{2} + \dots + \frac{y_{n-1} + y_n}{2} \right)$$

$$= \left[\Delta x \left(\frac{y_0}{2} + y_1 + y_2 + \dots + y_{n-1} + \frac{y_n}{2} \right) \right] = \left[\frac{1}{2} \text{left R-S} + \frac{1}{2} \text{right R-S} \right]$$

Trapezoidal Rule.

③ Simpson's rule. (needs n even)



AREA under parabola:

$$\underbrace{2\Delta x}_{\text{base}} \underbrace{\left(\frac{y_0 + 4y_1 + y_2}{6} \right)}_{\text{average}}$$

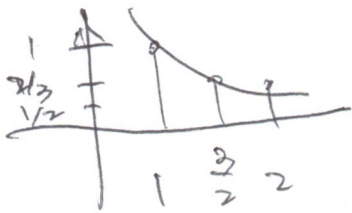
$$\frac{2\Delta x}{6} (y_0 + 4y_1 + y_2 + y_2 + 4y_3 + y_4 + \dots + y_{n-2} + 4y_{n-1} + y_n)$$

$$= \frac{\Delta x}{3} (y_0 + 4y_1 + 2y_2 + 4y_3 + \dots + 2y_{n-2} + 4y_{n-1} + y_n)$$

Lecture 25. Numerical Integration

$$\int_1^2 \frac{dx}{x} = \ln x \Big|_1^2 = \ln 2 - \ln 1 = \ln 2 \approx 0.693147$$

Two Intervals



① trapezoidal rule

$$\Delta x \left(\frac{1}{2} y_0 + y_1 + \frac{1}{2} y_2 \right)$$

$$\frac{1}{2} \left(\frac{1}{2} + 2/3 + \frac{1}{2} \cdot \frac{1}{2} \right)$$

$$\approx 0.76$$

② Simpson rule.

$$\frac{\Delta x}{3} (y_0 + 4y_1 + y_2)$$

$$= \frac{1}{6} \left(1 + \frac{8}{3} + \frac{1}{2} \right)$$

$$\approx 0.69444$$

Simpson's rule

$$\approx (\Delta x)^4$$

e.g. $\Delta x = \frac{1}{10}$

$$(\Delta x)^4 \approx 10^{-4}$$

Simpson's rule is derived using the exact answer for all degree 2 polynomials

(also exact for cubics)

Watch out for $\frac{1}{x}$ near 0

check = $f(x) = 1$ for these ① ② ③ rule.

Recall.

$V = \text{vol under}$



2D normal distribution?

e^{-r^2} rotated

$$V = \int_0^\infty \underbrace{2\pi r}_{\text{circ}} \underbrace{e^{-r^2}}_h \underbrace{dr}_{\text{thick}}$$

$$= -\pi e^{-r^2} \Big|_0^\infty = \pi(1-0) = \pi$$

$$Q = \int_{-\infty}^{\infty} e^{-t^2} dt$$



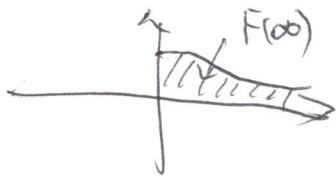
$Q = \text{area under } e^{-t^2}$

We will compute V by slices

$$V = Q^2; \Rightarrow Q = \sqrt{\pi}$$

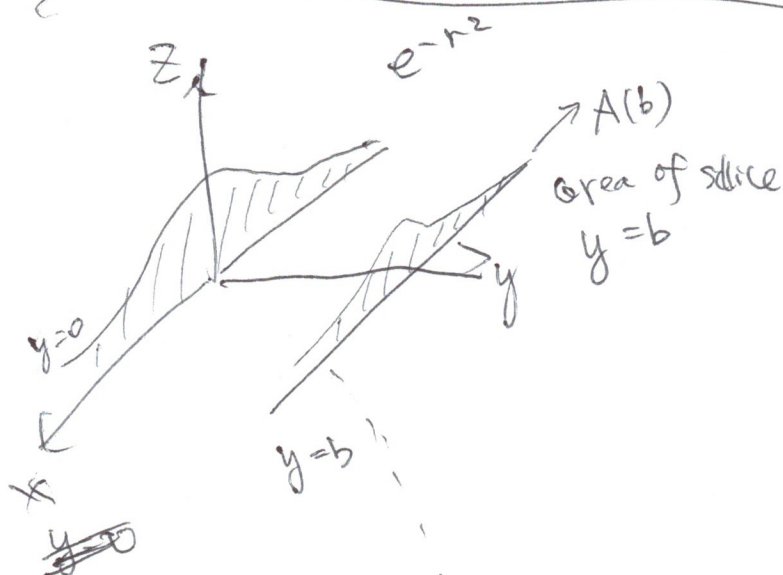
$$F(x) = \int_0^x e^{-t^2} dt$$

$$F(\infty) = \int_0^{\infty} e^{-t^2} dt$$



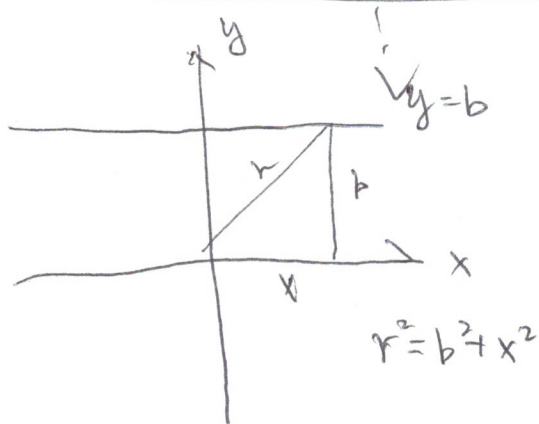
$$Q = 2F(\infty)$$

$$F(\infty) = \frac{\sqrt{\pi}}{2}$$



$$V = \int_{-\infty}^{\infty} A(y) dy$$

Fix $y=b$ and find $A(b)$



top view

$$e^{-r^2} = e^{-(b^2 + x^2)}$$

$$= e^{-b^2} \cdot e^{-x^2}$$

$A(b)$ area under ce^{-x^2}
 $c = e^{-b^2}$

$$A(b) = \int_{-\infty}^{\infty} e^{-b^2} e^{-x^2} dx$$

$$= e^{-b^2} \int_{-\infty}^{\infty} e^{-x^2} dx$$

$$A(b) = e^{-b^2} Q$$

$$V = \int_{-\infty}^{\infty} A(y) dy = \int_{-\infty}^{\infty} e^{-y^2} Q dy$$

$$= Q \int_{-\infty}^{\infty} e^{-y^2} dy$$

$$= Q Q = Q^2$$